

9 Electricity

9.1 Electric current

Candidates should be able to:

- 1 understand that an electric current is a flow of charge carriers
- 2 understand that the charge on charge carriers is quantised
- 3 recall and use $Q = It$
- 4 use, for a current-carrying conductor, the expression $I = Anvq$, where n is the number density of charge carriers

9.2 Potential difference and power

Candidates should be able to:

- 1 define the potential difference across a component as the energy transferred per unit charge
- 2 recall and use $V = W/Q$
- 3 recall and use $P = VI$, $P = I^2R$ and $P = V^2/R$

9.3 Resistance and resistivity

Candidates should be able to:

- 1 define resistance
- 2 recall and use $V = IR$
- 3 sketch the I - V characteristics of a metallic conductor at constant temperature, a semiconductor diode and a filament lamp
- 4 explain that the resistance of a filament lamp increases as current increases because its temperature increases
- 5 state Ohm's law
- 6 recall and use $R = \rho L/A$
- 7 understand that the resistance of a light-dependent resistor (LDR) decreases as the light intensity increases
- 8 understand that the resistance of a thermistor decreases as the temperature increases (it will be assumed that thermistors have a negative temperature coefficient)

10 D.C. circuits

10.1 Practical circuits

Candidates should be able to:

- 1 recall and use the circuit symbols shown in section 6 of this syllabus
- 2 draw and interpret circuit diagrams containing the circuit symbols shown in section 6 of this syllabus
- 3 define and use the electromotive force (e.m.f.) of a source as energy transferred per unit charge in driving charge around a complete circuit
- 4 distinguish between e.m.f. and potential difference (p.d.) in terms of energy considerations
- 5 understand the effects of the internal resistance of a source of e.m.f. on the terminal potential difference

10.2 Kirchhoff's laws

Candidates should be able to:

- 1 recall Kirchhoff's first law and understand that it is a consequence of conservation of charge
- 2 recall Kirchhoff's second law and understand that it is a consequence of conservation of energy
- 3 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in series
- 4 use the formula for the combined resistance of two or more resistors in series
- 5 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in parallel
- 6 use the formula for the combined resistance of two or more resistors in parallel
- 7 use Kirchhoff's laws to solve simple circuit problems

10.3 Potential dividers

Candidates should be able to:

- 1 understand the principle of a potential divider circuit
- 2 recall and use the principle of the potentiometer as a means of comparing potential differences
- 3 understand the use of a galvanometer in null methods
- 4 explain the use of thermistors and light-dependent resistors in potential dividers to provide a potential difference that is dependent on temperature and light intensity